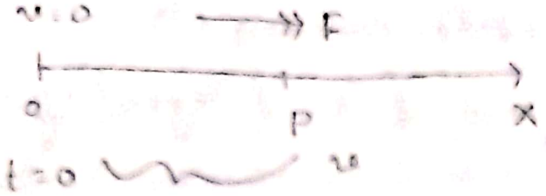


Let a particle starts from O and moves along OX. Let P be the position of the particle at any time t such that $OP = x$

Let v be the velocity of the particle at P. Then

$v = \frac{dx}{dt}$ = velocity of the particle at P (in the direction of x increasing)

$f = \frac{d^2x}{dt^2}$ = Acceleration of the particle at P (in the direction of x increasing)

$= \frac{dv}{dt} = v \cdot \frac{dv}{dx}$.

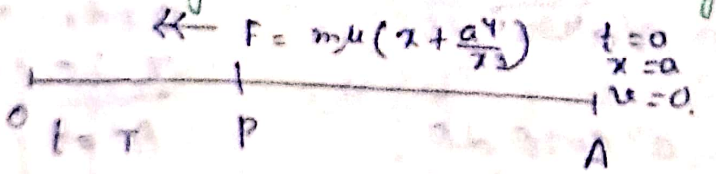
Equ^n of motion $\longrightarrow$

$m\frac{d^2x}{dt^2} = F$ [F is external force acting on the particle in the positive direction of x-axis]

$= -F$ [In the negative direction of x-axis]

Q. A particle of mass m is acted on by a force $m\mu\left(x + \frac{a^4}{x^3}\right)$ towards the origin where μ and a are constants and x is the distance of the particle from the origin. If the particle starts from rest at a distance a, find the time taken by it to arrive at origin.

Ans.



Let O be the origin and the particle starts from A and moves along AO. Let P be the position of the particle at any time t such that $OP = x$.

Let v be the velocity of the particle at P.

Equ^n of motion of the particle is $\longrightarrow$

$$mv\frac{dv}{dx} = -m\mu\left(x + \frac{a^4}{x^3}\right)$$

$$v\,dv = -\mu\left(x + \frac{a^4}{x^3}\right)dx$$

Integrating we get,

$$\frac{v^2}{2} = -\mu\left(\frac{x^2}{2} - \frac{a^4}{2x^2}\right) + c_1 \longrightarrow ①$$

[c_1 = integrating constant]

Initially $x = a$ and $v = 0$.

$$0 = -\mu\left(\frac{a^2}{2} - \frac{a^2}{2}\right) + c_1$$

$$c_1 = 0.$$

Equ^n ① becomes

$$\frac{v^2}{2} = -\mu\left(\frac{x^2}{2} - \frac{a^4}{2x^2}\right)$$

$$v^2 = -\mu\left(x^2 - \frac{a^4}{x^2}\right)$$

$$v^2 = \mu\left(\frac{a^4 - x^4}{x^2}\right)$$

$$v = -\sqrt{\mu}\,\frac{\sqrt{a^4 - x^4}}{x}$$ [As t increases x decreases]

$$\frac{dx}{dt} = -\sqrt{\mu}\,\frac{\sqrt{a^4 - x^4}}{x}$$

$$\sqrt{\mu}\,dt = -\frac{x\,dx}{\sqrt{a^4 - x^4}}$$

Let T be the time taken by the particle to arrive at origin. Integrating between the limits $t = 0$ when $x = a$ to $t = T$ when $x = 0$ we get

$$\int_0^T \sqrt{\mu}\,dt = -\int_a^0 \frac{x\,dx}{\sqrt{a^4 - x^4}}$$

$$\sqrt{\mu}\,T = \int_0^a \frac{x\,dx}{\sqrt{a^4 - x^4}}$$

Let $x^2 = a^2 \sin\theta$

$2x\,dx = a^2 \cos\theta\,d\theta$

x	0	a
θ	0	$\pi/2$

$$\sqrt{\mu}\,T = \frac{1}{2}\int_0^{\pi/2} \frac{a^2 \cos\theta\,d\theta}{a^2 \cos\theta}$$

$$= \frac{1}{2}\int_0^{\pi/2} d\theta$$

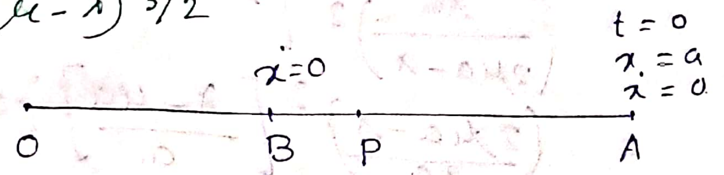
$$\sqrt{\mu}\,T = \frac{1}{2}\left[\theta\right]_0^{\pi/2}$$

$$T = \frac{\pi}{4\sqrt{\mu}}$$

This is the required time taken by the particle to arrive at origin.

Q. A particle moves in a st. line with an accelaration towards a fixed pt. in the st. line which is equal to $\left(\frac{\mu}{x^2} - \frac{\lambda}{x^3}\right)$ when the particle at a distance x from the given pt. it starts from rest at a distance a, show that it occillates between this distance and the distance $\frac{\lambda a}{2\mu a - \lambda}$ and it periodic time is $\frac{2\pi\mu a^3}{(2a\mu - \lambda)^{3/2}}$

Ans.



Let O be the fixed pt. Let the particle starts from A and moves along AO. such that $AO = a$. Let P be the position of the particle at any time t such that $OP = x$.

Equ^n of motion of the particle is

$$m\frac{d^2x}{dt^2} = -m\left(\frac{\mu}{x^2} - \frac{\lambda}{x^3}\right) \quad [m \text{ is mass of the particle}]$$

Multiplying by $2\dot{x}$ and integrating we get,

$$\left(\frac{dx}{dt}\right)^2 = \frac{2\mu}{x} - \frac{\lambda}{x^2} + C_1 \quad [C_1 = \text{integrating constant}]$$

Initially $x = a$, $\dot{x} = 0$

$$\frac{2\mu}{a} - \frac{\lambda}{a^2} + C_1 = 0.$$

$$C_1 = \frac{\lambda}{a^2} - \frac{2\mu}{a}$$

$$\therefore \left(\frac{dx}{dt}\right)^2 = \frac{2\mu}{x} - \frac{\lambda}{x^2} + \frac{\lambda}{a^2} - \frac{2\mu}{a}$$

$$\left(\frac{dx}{dt}\right)^2 = 2\mu\left(\frac{1}{x} - \frac{1}{a}\right) + \lambda\left(\frac{1}{a^2} - \frac{1}{x^2}\right)$$

$$= 2\mu\left(\frac{a-x}{ax}\right) - \lambda\left(\frac{a^2-x^2}{a^2x^2}\right)$$

$$\left(\frac{dx}{dt}\right)^2 = \frac{a-x}{a^2x^2}\left[2\mu ax - \lambda(a+x)\right] \longrightarrow (1)$$

The particle comes to rest when $\frac{dx}{dt} = 0$.

i.e, when $x = a$ and $x = \frac{\lambda a}{2\mu a - \lambda}$

Let B be the other pt. where the velocity of the particle again zero.

$$\therefore OB = \frac{\lambda a}{2\mu a - \lambda}$$

Force at the pt. A is $\frac{m}{a^3}(a\mu - \lambda) < 0$

i.e, $a\mu - \lambda < 0$

Force at the pt. B is $\frac{1}{\left(\frac{\lambda a}{2\mu a - \lambda}\right)^2}\left[\mu - \frac{\lambda(2\mu a - \lambda)}{\lambda a}\right]$

$$= \left(\frac{2\mu a - \lambda}{\lambda a}\right)^2 \left[\frac{\lambda - \mu a}{a}\right] > 0.$$

as $a\mu - \lambda < 0$.

Velocity at the pt. A and B are zero but force at the pt. A is not zero and directed towards O. Also force at the pt. B is not zero and directed away from the origin.

$\therefore$ The particle oscillate between the pts

$x = a$ and $x = \frac{\lambda a}{2\mu a - \lambda}$

Let $\frac{\lambda a}{2\mu a - \lambda} = p.a$ where $p = \frac{\lambda}{2\mu a - \lambda}$

$$\therefore \left(\frac{dx}{dt}\right)^2 = \frac{a-x}{a^\gamma x^2}\left[x(2\mu a - \lambda) - \lambda a\right]$$

$$= \frac{a-x}{a^\gamma x^2}\left[x.\frac{\lambda}{p} - \lambda a\right]$$

$$= \frac{\lambda}{p}\,\frac{a-x}{a^\gamma x^2}\left[x - ap\right]$$

$$\therefore \frac{dx}{dt} = -\sqrt{\frac{\lambda}{p}}\cdot\frac{\sqrt{(a-x)(x-ap)}}{ax}$$

[Negetive sign is taken as t increasing but x decreasing]

or, $\sqrt{\frac{\lambda}{p}}\,dt = -\frac{ax\,dx}{\sqrt{(a-x)(x-ap)}}$

If t_1 be the time taken by the particle in moving from $x = a$ to $x = pa$

Then we get,

$$\int_0^{t_1} \sqrt{\frac{\lambda}{p}}\, dt = -\int_a^{pa} \frac{ax\, dx}{\sqrt{(a-x)(x-ap)}}$$

Put $x = a\sin^2\theta + pa\cos^2\theta$

$dx = (2a\sin\theta\cos\theta - pa\, 2\cos\theta\sin\theta)\, d\theta$

$dx = 2a\sin\theta\cos\theta(1-p)\, d\theta$

x	pa	a
θ	0	$\pi/2$

$$\sqrt{\frac{\lambda}{p}}\, t_1 = -\int_{\pi/2}^{0} \frac{a(a\sin^2\theta + pa\cos^2\theta) \cdot 2a\sin\theta\cos\theta(1-p)\, d\theta}{\sqrt{a(1-p)\cos^2\theta\; a(1-p)\sin^2\theta}}$$

$$\sqrt{\frac{\lambda}{p}}\, t_1 = -2a^2\int_{\pi/2}^{0} (\sin^2\theta + p\cos^2\theta)\, d\theta$$

$$\sqrt{\frac{\lambda}{p}}\, t_1 = 2a^2\int_0^{\pi/2} (\sin^2\theta + p\cos^2\theta)\, d\theta$$

$$\sqrt{\frac{\lambda}{p}}\, t_1 = 2a^2\left[\frac{1}{2}\cdot\frac{\pi}{2} + \frac{p}{2}\cdot\frac{\pi}{2}\right]$$

$$\boxed{\int_0^{\pi/2} \sin^p\theta\cos^q\theta\, d\theta = \frac{\Gamma\frac{p+1}{2}\,\Gamma\frac{q+1}{2}}{2\,\Gamma\frac{p+q+2}{2}} \qquad \int_0^{\pi/2} \sin^2\theta\, d\theta = \frac{\Gamma\frac{2+1}{2}\,\Gamma\frac{1}{2}}{2\,\Gamma\frac{2+2}{2}} = \frac{\frac{1}{2}\pi}{2} = \frac{\pi}{4}}$$

$$\sqrt{\frac{\lambda}{p}}\, t_1 = \frac{\pi a^2}{2}\left[1 + \frac{\lambda}{2\mu a - \lambda}\right]$$

$$\sqrt{\frac{\lambda}{p}}\, t_1 = \pi a^3 \frac{\mu}{2\mu a - \lambda}$$

$$t_1 = \sqrt{\frac{p}{\lambda}}\, \pi a^3 \frac{\mu}{2\mu a - \lambda}$$

$$t_1 = \sqrt{\frac{1}{2\mu a - \lambda}}\, \frac{\pi a^3\mu}{2\mu a - \lambda}$$

$$t_1 = \frac{\pi a^3\mu}{(2\mu a - \lambda)^{3/2}}$$

Hence time of complete oscillation / periodic time is $2t_1 = \dfrac{2\pi a^3\mu}{(2\mu a - \lambda)^{3/2}}$

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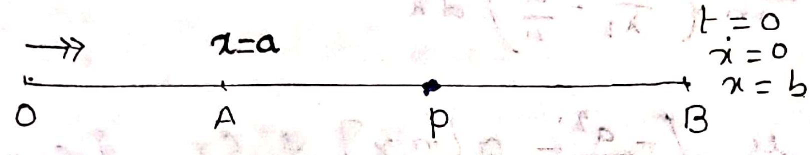
Q/ A particle starts from rest at a distance $b\,(>a)$ from a fixed pt. O under the action of a force through the fixed pt, the law of which at a distance x from O is $F = \mu\left(1-\frac{a}{x}\right)$ towards O when $x>a$ but $F=\mu\left(\frac{a^2}{x^2}-\frac{a}{x}\right)$ away from O when $x<a$. Show that the velocity of the particle will again vanish when $x=\frac{a^2}{b}$ [2001]

Ans.



Let the fixed pt. be O. Let the particle starts from B such that $OB=b$ and let A be a pt. on OB such that $OA = a\,(<b)$. Let us consider the motion from B to A.

Equ^n of motion will be

$$m\frac{d^2x}{dt^2} = -\mu\left(1-\frac{a}{x}\right)$$

$$v\frac{dv}{dx} = -\frac{\mu}{m}\left(1-\frac{a}{x}\right)$$

$$v\,dv = -\frac{\mu}{m}\left(1-\frac{a}{x}\right)dx$$

Integrating $\rightarrow$

$$\frac{v^2}{2} = -\frac{\mu}{m}\left(x-a\log x\right)+C_1 \qquad [C_1 = \text{Integrating constant}]$$

Initially when $t=0$, $x=b$, $\dot{x}=0$.

$$0 = -\frac{\mu}{m}\left(b-a\log b\right)+C_1$$

$$C_1 = \frac{\mu}{m}\left(b-a\log b\right)$$

$$\therefore\ v^2 = -\frac{2\mu}{m}\left(x-a\log x\right)+\frac{2\mu}{m}\left(b-a\log b\right)$$

$$v^2 = \frac{2\mu}{m}\left(b-a\log b-x+a\log x\right) \longrightarrow (1)$$

Let V be the velocity of the particle when it reaches A

i.e, $x=a$

$$V^2 = \frac{2\mu}{m}\left(b-a\log b-a+a\log a\right) \longrightarrow (2)$$

When the particle crosses the pt. A and moves towards O i.e, when $x < a$, the equn of motion will be

$$m\frac{d^2x}{dt^2} = \mu\left(\frac{a^2}{x^2} - \frac{a}{x}\right)$$ [We take positive sign since the law of force is away from O]

$$v\frac{dv}{dx} = \frac{\mu}{m}\left(\frac{a^2}{x^2} - \frac{a}{x}\right)$$

$$v.dv = \frac{\mu}{m}\left(\frac{a^2}{x^2} - \frac{a}{x}\right)dx$$

Integrating

$$\frac{v^2}{2} = \frac{\mu}{m}\left[-\frac{a^2}{x} - a\log x\right] + c_2$$ [c_2 = Integrating Constant]

at A, $x = a$. $v = V$

$$\frac{V^2}{2} = \frac{\mu}{m}\left[-a - a\log a\right] + c_2$$

$$c_2 = \frac{V^2}{2} + \frac{\mu}{m}(a + a\log a)$$

$$\therefore \frac{v^2}{2} = \frac{\mu}{m}\left[-\frac{a^2}{x} - a\log x\right] + \frac{V^2}{2} + \frac{\mu}{m}(a + a\log a)$$

$$v^2 = \frac{2\mu}{m}\left[a + a\log a - \frac{a^2}{x} - a\log x\right] + V^2$$

Let the particle again comes to rest at C where OC=d

$\therefore x = d$, $v = 0$.

$$V^2 + \frac{2\mu}{m}\left[a + a\log a - \frac{a^2}{d} - a\log d\right] = 0$$

$$\frac{2\mu}{m}\left[b - a\log b - a + a\log a\right] + \frac{2\mu}{m}\left[a + a\log a - \frac{a^2}{d} - a\log d\right] = 0$$

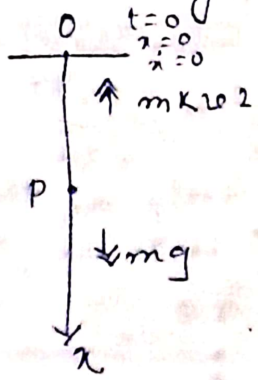
$$b - a\log b - \cancel{a} + a\log a + \cancel{a} + a\log a - \frac{a^2}{d} - a\log d = 0 \quad (4)$$

We see that $d = \frac{a^2}{b}$ satisfies the equn (4)

Hence the velocity of the particle will again vanish when $x = \frac{a^2}{b}$.

Q A particle falls under constant gravity through a distance x starting from rest. A small resistance per unit mass equal to k times the square of the speed acts on the particle. Show that the Kinetic Energy acquired is approximately equal to $mgx(1-kx)$ neglecting highest power of x. $\boxed{2001}$

Ans. Let the particle starts from O and moves along OX vertically downwards. Let P be the position of the particle at any time t such that $OP = x$.
Also let v be the velocity of the particle at P.



Equ^n of motion of the particle is.

$$mv\frac{dv}{dx} = mg - mkv^2$$

$$v\frac{dv}{dx} = g - kv^2$$

$$\frac{1}{2}\frac{d}{dx}(v^2) + kv^2 = g.$$

$$\frac{d}{dx}(v^2) + 2kv^2 = 2g$$

This is a linear equ^n in v^2

$$\therefore I.F = e^{\int 2k\,dx} = e^{2kx}$$

Multiplying by I.F and integrating

$$v^2 e^{2kx} = 2g\int e^{2kx}\,dx$$

$$v^2 e^{2kx} = 2g\frac{e^{2kx}}{2k} + C_1$$

Initially when $t=0$, $x=0$, $\dot{x}=0$.

$$C_1 = -\frac{g}{k}.$$

$$\therefore v^2 e^{2kx} = \frac{g}{k}(e^{2kx} - 1)$$

$$v^2 = \frac{g}{k} - \frac{g}{k}e^{-2kx}.$$

$$v^2 = \frac{g}{k} - \frac{g}{k}\left[1 - 2kx + \frac{(2kx)^2}{2!} - \cdots\right]$$

$$v^2 = \frac{g}{k} - \frac{g}{k} + 2gx - 2kx^2 g.$$

$$v^2 = 2gx - 2gkx^2$$

$$v^2 = 2gx[1 - kx]$$

$$K.E = \frac{1}{2}mv^2$$

$$K.E = \frac{1}{2}m\,2gx(1-kx)$$

$$K.E = mgx(1-kx) \quad \text{(proved.)}$$

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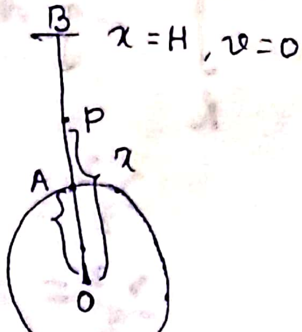
Q. A particle is projected from the earth surface vertically upwards to the velocity P. If h and H be the greatest height attained by the particle moving on uniform and variable acceleration respectively. Show that $\frac{1}{h} - \frac{1}{H} = \frac{1}{R}$. Where R is the radius of the earth.



Ans. Let O be the centre of the earth, which is choosen as origin. Let the particle be projected from A and P be the position of particle at any time t, such that $OP = x$

Case I → When the acceleration is not uniform, it is one of inverse square to the centre of earth. So if 'f' be the accelaration of the particle at P, then $f = -\frac{\mu}{x^2}$

But on the surface of the earth $f = -g$.

$\therefore \mu = gR^2$

Equ^n of motion of the particle is $v\frac{dv}{dx} = -\frac{\mu}{x^2}$ [where v be the velocity of the particle at P]

$$v\frac{dv}{dx} = -\frac{\mu}{x^2} \longrightarrow (1)$$

Since, H is the greatest height attained by the particle, when $x = H$, $\dot{x} = v = 0$.

Integrating (1) between the limits $x = R$ and $v = V$ to $x = H$ when $v = 0$ we get

$$\int_R^{H+R} -\frac{\mu}{x^2}\,dx = \int_V^0 v\,dv$$

$$\frac{V^2}{2} = \left[\frac{\mu}{x}\right]_R^{H+R}$$

$$\frac{V^2}{2} = \frac{\mu}{R} - \frac{\mu}{H+R}$$

$$\frac{V^2}{2} = \mu\left(\frac{1}{R} - \frac{1}{H+R}\right) \longrightarrow (2)$$

Case II → When acceleration is uniform. in this case equ^n of motion is

$$v\frac{dv}{dx} = -g \longrightarrow (3)$$

Since h is the greatest height attained by the particle, when $x = h$, $\dot{x} = v = 0$

Integrating ③ between the limits $x = R$ and $v = V$ to $x = h$ to $v = 0$ we get,

$$\int_R^{R+h} -g\,dx = \int_V^0 v\,dv$$

$$\frac{V^2}{2} = g(R+h-R)$$

$$\frac{V^2}{2} = gh \longrightarrow (4)$$

Eliminating $\frac{V^2}{2}$ from ② and ④

$$\mu\left(\frac{1}{R} - \frac{1}{H+R}\right) = gh$$

$$gR^2\left(\frac{1}{R} - \frac{1}{H+R}\right) = gh$$

$$\text{or, } R^2\left(\frac{H+R-R}{R(H+R)}\right) = h$$

$$\text{or, } RH = h(H+R)$$

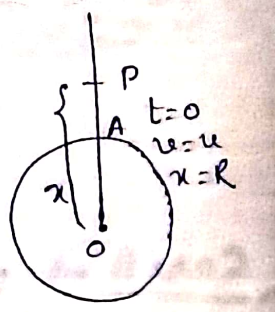
$$\text{or, } RH = hH + hR.$$

$$\text{or, } \frac{1}{h} = \frac{1}{R} + \frac{1}{H}$$

$$\text{or } \frac{1}{h} - \frac{1}{H} = \frac{1}{R}. \text{ (proved)}$$

9. A particle is projected vertically upwards from the earth surface with a velocity just sufficient to carry it to infinity. Prove that the time is take in reaching a height h is $\frac{1}{3}\sqrt{\frac{2R}{g}}\left[\left(1+\frac{h}{R}\right)^{3/2} - 1\right]$

Ans. Let O be the centre of the earth, which is chosen as origin. Let the particle be projected from A and P be the position of the particle at any time 't'. Let v be the velocity at P. Equation of motion of the particle is



$$m v \frac{dv}{dx} = -m\frac{\mu}{x^2} \longrightarrow (1) \text{ [m is the mass of the particle]}$$

$$\text{or, } \int_u^0 v\,dv = \int_R^\infty -\frac{\mu}{x^2}\,dx$$ [where u is the velocity of projection of particle just sufficient to carry it to a infinite distance]

or, $\frac{u^2}{2} = \frac{\mu}{R} \longrightarrow (2)$

On the surface of the earth

$$\frac{\mu}{R^2} = g$$

$$\mu = gR^2$$

From (2)

$$\frac{u^2}{2} = gR$$

$$u = \sqrt{2gR} \longrightarrow (3)$$

Equ^n (1) can be written as,

$$v\frac{dv}{dx} = -\frac{\mu}{x^2}$$

$$v\,dv = -\frac{\mu}{x^2}dx$$

or, $\frac{v^2}{2} = \frac{\mu}{x} + C_1$

Initially $x = R$, $v = \sqrt{2gR}$.

$$C_1 = gR - \frac{\mu}{R}.$$

$$C_1 = gR - gR = 0.$$

$$\therefore v^2 = 2\frac{gR^2}{x}$$

$$v = \sqrt{2g}\frac{R}{\sqrt{x}}$$ [positive sign is taken as t increases x increases]

or, $\frac{dx}{dt} = \sqrt{2g}\frac{R}{\sqrt{x}}.$

$$\sqrt{x}\,dx = \sqrt{2g}\,R\,dt$$

Let t_1 be the time taken by the particle to reach the greatest height above the surface of the earth.

Then $t = t_1$, $x = R+h$.

Integrating between the limits $t = 0$ when $x = R$ to $t = t_1$ when $x = R+h$ we get

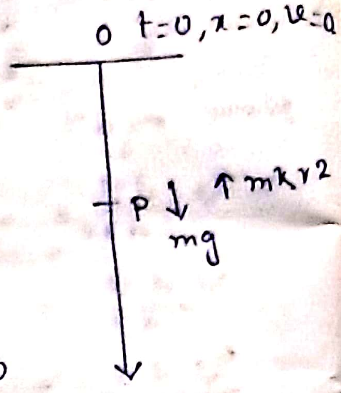
$$\int_R^{R+h}\sqrt{x}\,dx = \sqrt{2g}R\int_0^{t_1}dt$$

$$\sqrt{2g}Rt_1 = \frac{2}{3}\left[x^{3/2}\right]_R^{R+h}$$

$$\sqrt{2g}Rt_1 = \frac{2}{3}\left[(R+h)^{3/2} - R^{3/2}\right]$$

Q. A particle falls under gravity (suppose constant) in a resisting medium whose resistance varies as the square of the velocity, To find the motion of the particle if it starts from rest. $\boxed{1996}$

Ans. Let particle starts from 'O' moves along OX vertically downwards.



Let P be the position of the particle at any time t, such that $OP = x$.

Also let v be the velocity of the particle P.

The forces acting on the particle P are shown in the figure.

Equ^n of motion of particle is,

$$mv\frac{dv}{dx} = mg - mkv^2$$

$$v\frac{dv}{dx} + kv^2 = g \longrightarrow (1)$$

Let V be the terminal velocity then

$$v\frac{dv}{dx} = 0 \quad \text{when } v = V$$

$\therefore$ From (1) $\quad kV^2 = g.$

$$k = \frac{g}{V^2}.$$

From (1) $\quad v\frac{dv}{dx} = g - \frac{g}{V^2}v^2.$

$$v\frac{dv}{dx} = \frac{g}{V^2}(V^2 - v^2)$$

$$\frac{v\,dv}{V^2 - v^2} = \frac{g}{V^2}dx.$$

Integrating

$$-\frac{1}{2}\log(V^2 - v^2) = \frac{g}{V^2}x + C_1 \longrightarrow (2)$$

Initially when $t=0, x=0, v=0.$

$$\therefore -\frac{1}{2}\log V^2 = C_1$$

From (2)

$$-\frac{1}{2}\log(V^2 - v^2) = \frac{g}{V^2}x - \frac{1}{2}\log V^2.$$

$$\log\frac{V^2}{V^2 - v^2} = \frac{2g}{V^2}x$$

$$\log \frac{V^2 - v^2}{V^2} = -\frac{2gx}{V^2}$$

$$\frac{V^2 - v^2}{V^2} = e^{-\frac{2gx}{V^2}}$$

$$1 - \frac{v^2}{V^2} = e^{-\frac{2gx}{V^2}}$$

$$v^2 = V^2\left(1 - e^{-\frac{2gx}{V^2}}\right) \longrightarrow (3)$$

This gives the velocity at any position.

Equ^n (1) can be written as

$$\frac{dv}{dt} = g\left(1 - \frac{v^2}{V^2}\right)$$

$$\frac{dv}{V^2 - v^2} = \frac{g}{V^2}\,dt$$

Integrating,

$$\frac{1}{V}\tanh^{-1}\frac{v}{V} = \frac{gt}{V^2} + C_2$$

~~Integrating between the limits x = 0 when v = xv for x = x1, v = 0 we get,~~

Initially, $t = 0,\ x = 0,\ v = u$

$$\therefore C_2 = 0.$$

$$\therefore \frac{1}{V}\tanh^{-1}\frac{v}{V} = \frac{gt}{V^2}.$$

$$\tanh^{-1}\frac{v}{V} = \frac{gt}{V}.$$

$$v = V\tanh\left(\frac{gt}{V}\right) \longrightarrow (4)$$

This gives the relation between velocity and time.

Equating v from (3) and (4) we get,

$$V^2\tanh^2\left(\frac{gt}{V}\right) = V^2\left(1 - e^{-\frac{2gx}{V^2}}\right)$$

$$\tanh^2\left(\frac{gt}{V}\right) = 1 - e^{-\frac{2gx}{V^2}}$$

$$e^{-\frac{2gx}{V^2}} = 1 - \tanh^2\left(\frac{gt}{V}\right)$$

$$e^{-\frac{2gx}{V^2}} = \operatorname{sech}^2\left(\frac{gt}{V}\right)$$

$$e^{\frac{2gx}{V^2}} = \cosh^2\left(\frac{gt}{V}\right)$$

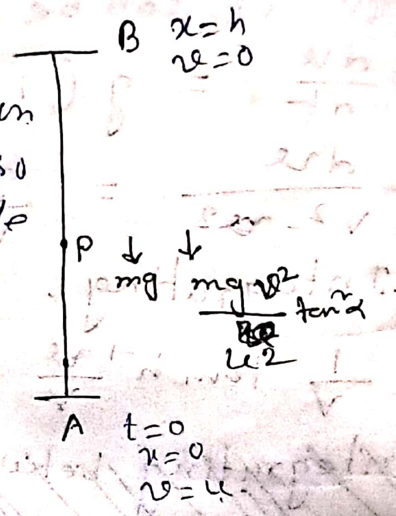
$$\frac{2gx}{V^2} = \log\left\{\cosh^2\left(\frac{gt}{V}\right)\right\}$$

$$x = \frac{V^2}{2g}\log\left\{\cosh^2\left(\frac{gt}{V}\right)\right\}$$

This gives the relation between x and t

Q. A heavy particle is projected upwards with a velocity u in a medium, the resistance of which is $\frac{g}{u^2}\tan^2\alpha$ times the square of the velocity, α being a constant. Show that the particle will return to the point of projection with a velocity $u\cos\alpha$, after a time $\frac{u}{g}\cot\alpha\left[\alpha+\log\frac{\cos\alpha}{1-\sin\alpha}\right]$

Ans. Let the particle will be projected from the pt. A and P be the position of the particle at any time t. Also let v be the velocity of the particle at P.



Equ^n of motion of the particle is

$$mv\frac{dv}{dx} = -mg - \frac{mgv^2}{u^2}\tan^2\alpha$$

$$v\frac{dv}{dx} = -g - \frac{g}{u^2}v^2\tan^2\alpha \longrightarrow (1)$$

or,

$$\frac{2v\tan^2\alpha\, dv}{u^2+v^2\tan^2\alpha} = -2\tan^2\alpha\frac{g}{u^2}dx$$

On integrating

$$\log(u^2+v^2\tan^2\alpha) = -2\tan^2\alpha\frac{g}{u^2}x + C_1$$

Initially when $t=0$, $x=0$, $v=u$

$$\therefore C_1 = \log(u^2+u^2\tan^2\alpha)$$

$$C_1 = \log u^2\sec^2\alpha.$$

$$\therefore \log(u^2+v^2\tan^2\alpha) = -\frac{2gx}{u^2}\tan^2\alpha + \log u^2\sec^2\alpha$$

or,

$$\frac{2gx}{u^2}\tan^2\alpha = \log(u^2\sec^2\alpha) - \log(u^2+v^2\tan^2\alpha) \longrightarrow (2)$$

If h is the maximum height attained by the particle then, when $x=h$, $v=0$.

From ② we have.

$$\frac{2gh}{u^2}\tan^2\alpha = \log u^2\sec^2\alpha - \log u^2.$$

$$\frac{2gh}{u^2}\tan^2\alpha = \log \sec^2\alpha.$$

$$\boxed{h = \frac{u^2}{2g}\cot^2\alpha \log\sec^2\alpha} \longrightarrow ③$$

Equⁿ ① can be written as

$$dt = -\frac{u^2}{g}\cdot\frac{dv}{u^2+v^2\tan^2\alpha} \longrightarrow (4).$$

let t_1 be the time taken by the particle to reach the heigest pt. B.

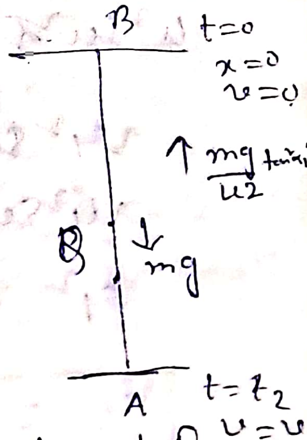
Integrating ④ between the limits $t=0$ to $t=t_1$ and $v=u$ to $v=0$, we get,

$$\int_0^{t_1} dt = -\frac{u^2}{g}\int_u^0 \frac{dv}{u^2+v^2\tan^2\alpha}$$

$$t_1 = \frac{1}{u}\frac{u^2}{g}\frac{1}{\tan\alpha}\left[\tan^{-1}\left(\frac{v\tan\alpha}{u}\right)\right]_0^u$$

$$t_1 = \frac{u}{g}\frac{1}{\tan\alpha}[\alpha]$$

$$t_1 = \frac{u\alpha}{g\tan\alpha} \longrightarrow ⑤.$$



- From the heigest pt. B, the particle starts moving downwards from rest.

Let v be the velocity of the particle at Q such that $BQ = x$.

Equⁿ of motion in downwards direction is

$$mv\frac{dv}{dx} = mg - \frac{mg}{u^2}v^2\tan^2\alpha \longrightarrow ⑥$$

$$\text{or}\quad \frac{-2v\,dv}{u^2 - v^2\tan^2\alpha} = -\frac{2g}{u^2}dx$$

Integrating,

$$\frac{1}{\tan^2\alpha}\log\left(u^2-v^2\tan^2\alpha\right) = -\frac{2gx}{u^2}+c_2$$

Initially $t=0,\ x=0,\ v=0$.

$$c_2 = \cot^2\alpha \log(u^2)$$

$$\therefore \frac{1}{\tan^2\alpha}\log\left(u^2-v^2\tan^2\alpha\right) = -\frac{2gx}{u^2}+\cot^2\alpha\log(u^2)$$

$$\frac{2gx}{u^2} = \cot^2\alpha\left[\log(u^2)-\log\left(u^2-v^2\tan^2\alpha\right)\right] \longrightarrow (7)$$

Let v_1 be the velocity of return. i.e, when $x=h,\ v=v_1$.

$\therefore$ From (7)

$$\frac{2gh}{u^2} = \cot^2\alpha\left[\log u^2-\log\left(u^2-v_1^2\tan^2\alpha\right)\right]$$

$$\frac{\cancel{2g}}{\cancel{u^2}}\,\frac{\cancel{u^2}}{\cancel{2g}}\cot^2\alpha\log\sec^2\alpha = \cot^2\alpha\left[\log\frac{u^2}{u^2-v_1^2\tan^2\alpha}\right]$$

$$\sec^2\alpha = \frac{u^2}{u^2-v_1^2\tan^2\alpha}$$

$$u^2\sec^2\alpha - v_1^2\sec^2\alpha\tan^2\alpha = u^2.$$

$$v_1^2\sec^2\alpha\tan^2\alpha = u^2(\sec^2\alpha-1)$$

$$v_1^2\sec^2\alpha\tan^2\alpha = u^2\tan^2\alpha.$$

$$v_1^2 = \frac{u^2}{\sec^2\alpha} = u^2\cos^2\alpha.$$

$$\boxed{v_1 = u\cos\alpha.} \longrightarrow (8)$$

This gives the required velocity when the particle comes back to the pt. of projection A.

Equ^n (6) can be written as,

$$\frac{dv}{dt} = \frac{g}{u^2}\left(u^2 - v^2 \tan^2\alpha\right)$$

$$dt = \frac{u^2}{g} \frac{dv}{u^2 - v^2 \tan^2\alpha} \longrightarrow (9)$$

Let t_2 be the time taken by the particle to reach the pt. A.

Integrating (9) between the limits $t=0$, when $v=0$, $t=t_2$ when $v=v_1$, we get,

$$\int_0^{t_2} dt = \frac{u^2}{g}\int_0^{v_1} \frac{dv}{u^2 - v^2 \tan^2\alpha}$$

Integrating

$$t_2 = \frac{u^2}{g} \cdot \frac{1}{2u\tan\alpha}\left[\log \frac{u + v\tan\alpha}{u - v\tan\alpha}\right]_0^{v_1}$$

$$t_2 = \frac{u}{2g\tan\alpha}\left[\log \frac{u + v_1\tan\alpha}{u - v_1\tan\alpha} - \log 1\right]$$

$$t_2 = \frac{u}{2g\tan\alpha}\left[\log \frac{u + u\cot\alpha\tan\alpha}{u - u\cos\alpha\tan\alpha}\right]$$

$$t_2 = \frac{u}{2g\tan\alpha}\log\left(\frac{1+\sin\alpha}{1-\sin\alpha}\right)$$

$$t_2 = \frac{u}{2g\tan\alpha}\left[\log \cancel{\frac{(1+\sin\alpha)^2}{\cos^2\alpha}} \frac{\cos^2\alpha}{(1-\sin\alpha)^2}\right]$$

$$t_2 = \frac{u}{g\tan\alpha}\log\frac{\cos\alpha}{1-\sin\alpha}$$

$\therefore$ The required time is $t_1 + t_2$

$$= \frac{u}{g\tan\alpha}\left[\alpha + \log\frac{\cos\alpha}{1-\sin\alpha}\right]$$

$$= \frac{u}{g}\cot\alpha\left[\alpha + \log\frac{\cos\alpha}{1-\sin\alpha}\right]$$